L2GTX: From Local to Global Time Series Explanations

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THE CONTEXT

Why Explainability Matters

Deep learning models achieve **impressive accuracy** in time series classification, but they operate as **opaque black boxes** .

We need to understand why the model makes a prediction, not just what it predicts.

Applications of Time Series Classification

Healthcare
ECG analysis, patient monitoring

Finance
Market prediction, risk assessment

Safety-Critical
Fault detection, anomaly monitoring

In high-stakes applications, explanations help validate predictions, build trust, and support decision-making.

Three Critical Gaps

1

Many model-agnostic XAI methods treat time steps as independent features, **overlooking temporal dependencies** [1,2].

2

Most time-series XAI methods provide local explanations, while class-wise global explanations **remain limited** [3-5].

3

Existing global approaches are often **model-specific** or rely on **surrogate-rule extraction** [6-8].

?

How can we derive **class-wise global explanations** that preserve **temporal structure** while remaining **model-agnostic**?

[1] Schlegel et al., 2019 [2] Schlegel et al., 2021 [3] Rojat et al., 2021 [4] Di et al., 2022
[5] Theissler et al., 2022 [6] Oviedo et al., 2019 [7] Siddiqui et al., 2019 [8] Mekonnen et al., 2023

L2GTX OVERVIEW

- Five-step process from local explanations to global insights.

1. Local Attribution → 2. Merge Clusters → 3. Global Importance → 4. Select Instances → 5. Aggregate Events

Input

- Time series instances + black-box model f + Budget B

Output

- Class-wise global explanations

Step 1: Local Explanations (LOMATCE)

- Generate local explanations using LOMATCE [9].

Generate Neighbors

- Perturb input
- Compute similarity weights

→

PEP Transformation

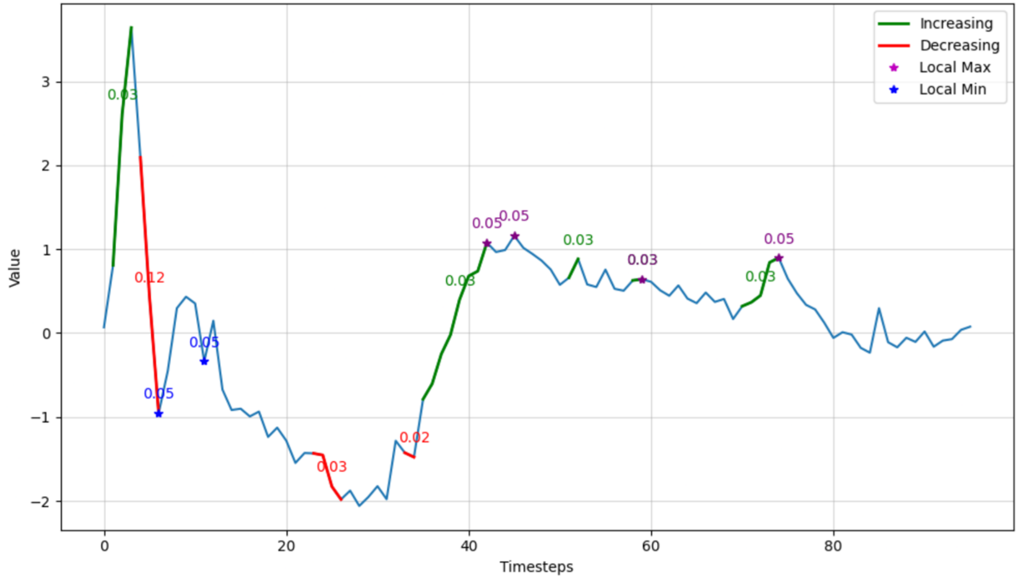
- Extract events (PEPs)
- Cluster similar events
- Build event representation

→

Train Linear Model

- Learn importance of event clusters

➢ **Parameterised Event Primitives (PEPs)** are domain-specific events characterised by parameters.



PEPs

Capture temporal patterns as human-understandable events:

Increasing (start, duration, gradient)	**Decreasing** (start, duration, gradient)
Local Max (time, value)	**Local Min** (time, value)

[9] Mekonnen et al., 2025

Step 2: Merge Similar Clusters

- Top clusters with importance are obtained per instance.

Instance-level clusters

Instance $_1$	Instance$_2$
Inc$_1$	Inc$_2$
Dec$_1$	Dec$_2$
Max$_1$	Max$_3$

→ Merge Similar clusters (threshold-controlled) →

Global clusters

(Inc_G$_1$) ←	Inc$_1$, inc$_2$
(Dec_G$_1$) ←	Dec$_1$, Dec$_2$
Max_G$_1$ ←	Max$_1$
Max_G$_2$ ←	Max $_3$

- Similar clusters are merged using hierarchical clustering.
- Merging is controlled by a percentile-based threshold.

Step 3: Global Cluster Importance

- Aggregate across instances → Global importance per cluster.

	Inc_G_1	Dec_G_2	Max_G_3
$Inst_1$	0.8	0	0.3
$Inst_2$	0	0.6	0.2
$Inst_3$	0.7	0.5	0

$$I_j = \sqrt{\sum_{i=1}^{N} |M_{i,j}|}, \quad j = 1, \ldots, G$$

→ Global Cluster Importance

N: *instances*
G: *global clusters*
$M_{i,j}$: *cluster importance* j *for instance* i

Step 4: Select Representative Instances

- Greedily maximise additional coverage.

All Instances

Inst_1	**Inst_2**	**Inst_3**	**Inst_4**	**Inst_5**
G_1, G_3	G_2, G_3	G_1	G_3	G_3, G_4

Submodular Pick [10]

Selected (B=2)

Inst_1

G_1, G_2

Inst_2

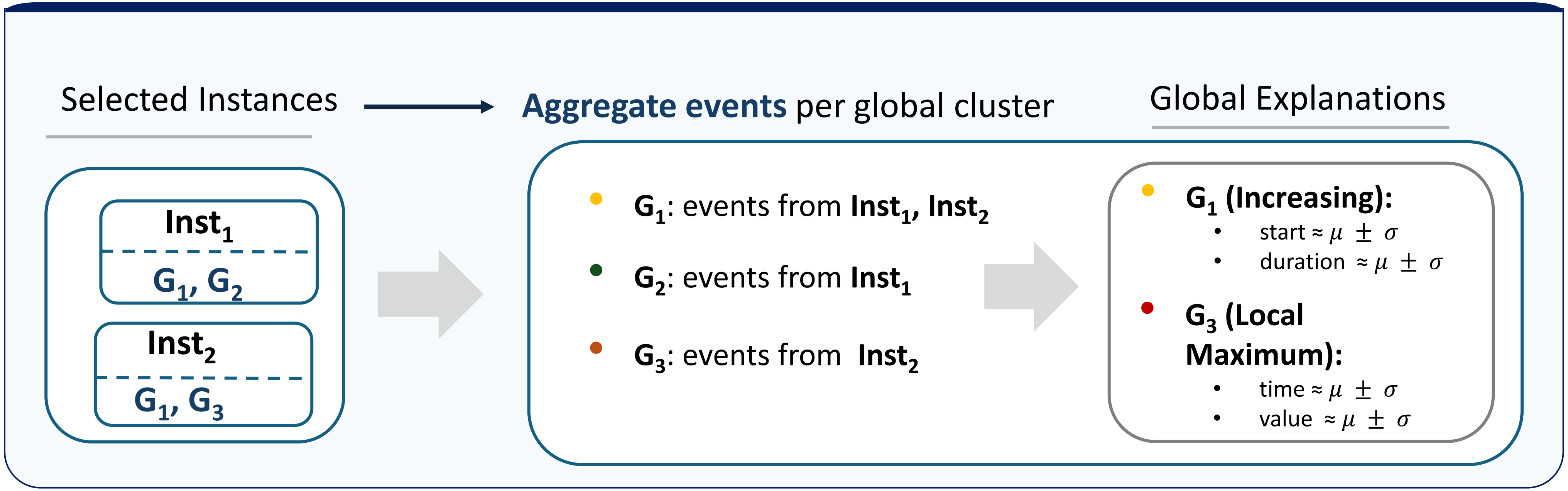
G_2, G_3

Covered Clusters: G_1, G_2, G_3

- Maximise coverage of important clusters.
- Avoid selecting instances that explain the same clusters.
- Stop when all important clusters are covered (or budget B is reached).

[10] Ribeiro et al., 2016

Step 5: Aggregate Events Into Global Explanations



- Events assigned to the same global cluster are aggregated and summarised using $mean \pm standard\ deviation$ to produce class-wise explanations.

What L2GTX Produces

Class-Wise Global Explanations

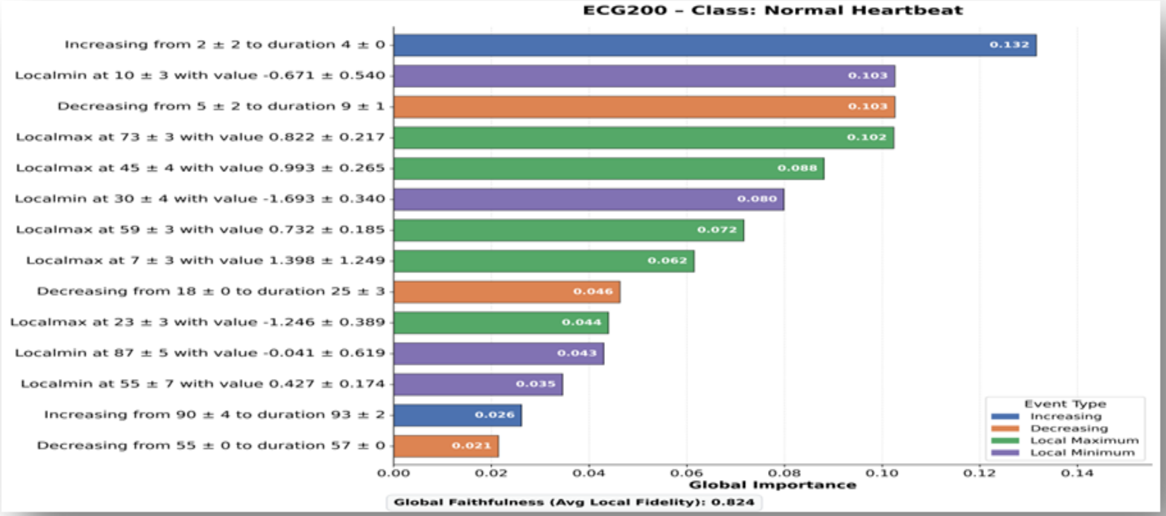
Bar Chart Format
Importance scores of aggregated event clusters

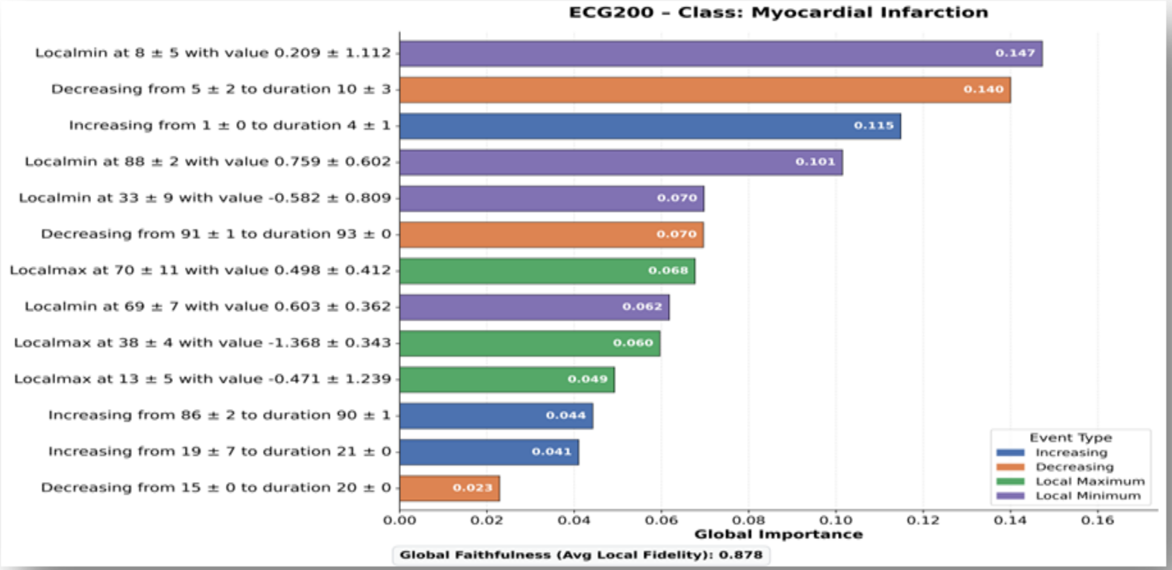
Colour Coding
Different colours for event types (trends vs extrema)

Temporal Statistics
Mean ± std for timing, duration, magnitude

What You Learn

WHAT patterns matter (trend/extrema)
WHERE they occur (temporal location)
HOW they manifest (duration, magnitude)
HOW MUCH they contribute (importance)





Black-Box Predictive Performance

- FCN and LSTM-FCN across six benchmark time series datasets.

6 Benchmark Datasets
Varying domains and difficulties

2 Black-Box Models
FCN • LSTM-FCN

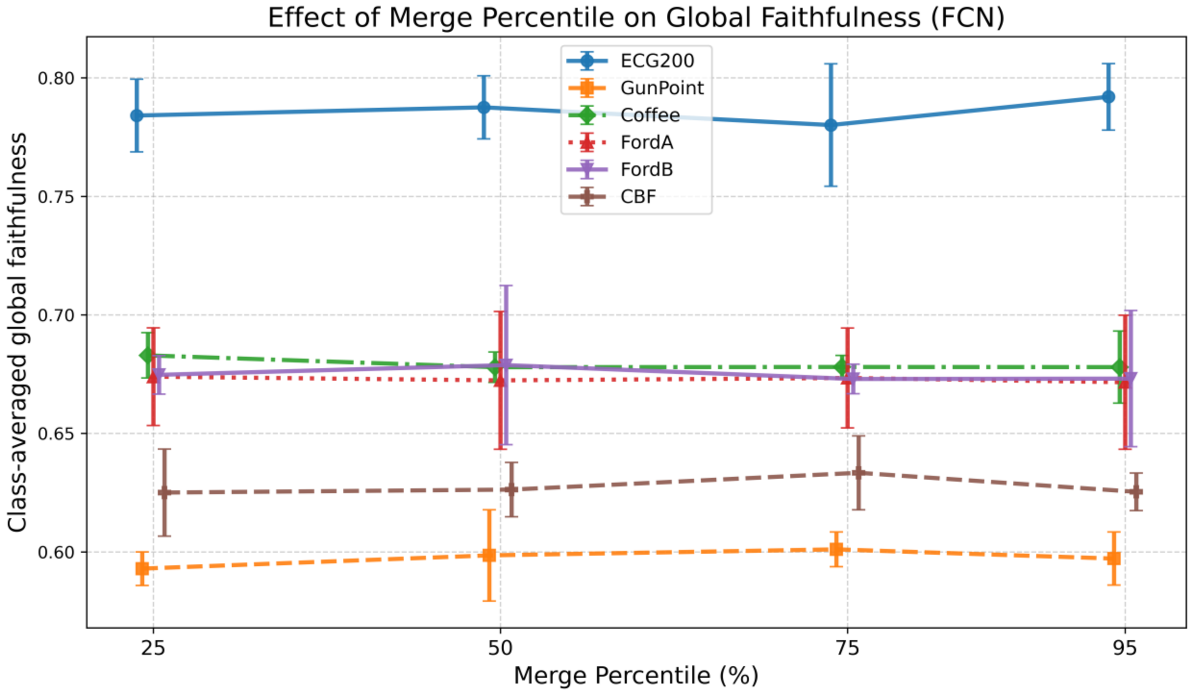
100 Repeated Runs
Random train/val/test splits
Mean accuracy ± 95% CI

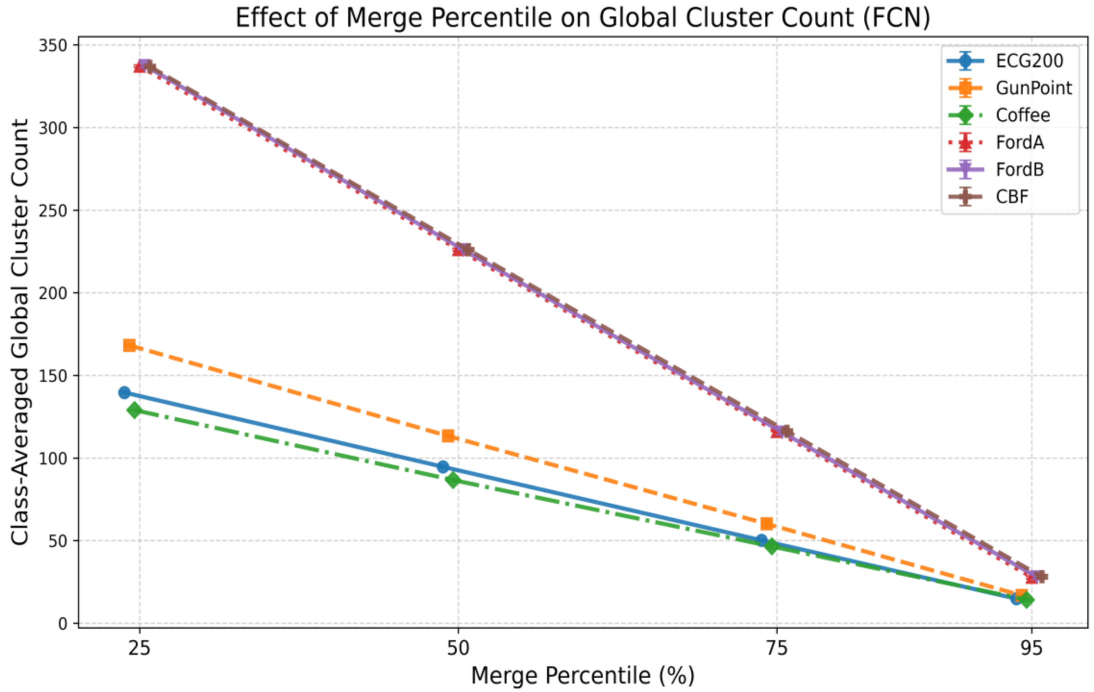
	Mean Test Accuracy (± 95% Confidence Interval)	
Dataset	**FCN**	**LSTM_FCN**
ECG200	0.87 ± 0.03	0.86 ± 0.01
GunPoint	0.99 ± 0.01	0.98 ± 0.01
Coffee	1.00 ± 0.00	1.00 ± 0.00
FordA	0.90 ± 0.02	0.88 ± 0.01
FordB	0.88 ± 0.02	0.86 ± 0.01
CBF	0.98 ± 0.01	1.00 ± 0.00

- Both models achieve high accuracy across all datasets, making them reliable targets for explanation.

Merge Percentile Sensitivity Analysis

- Compact global explanations without loss of faithfulness.

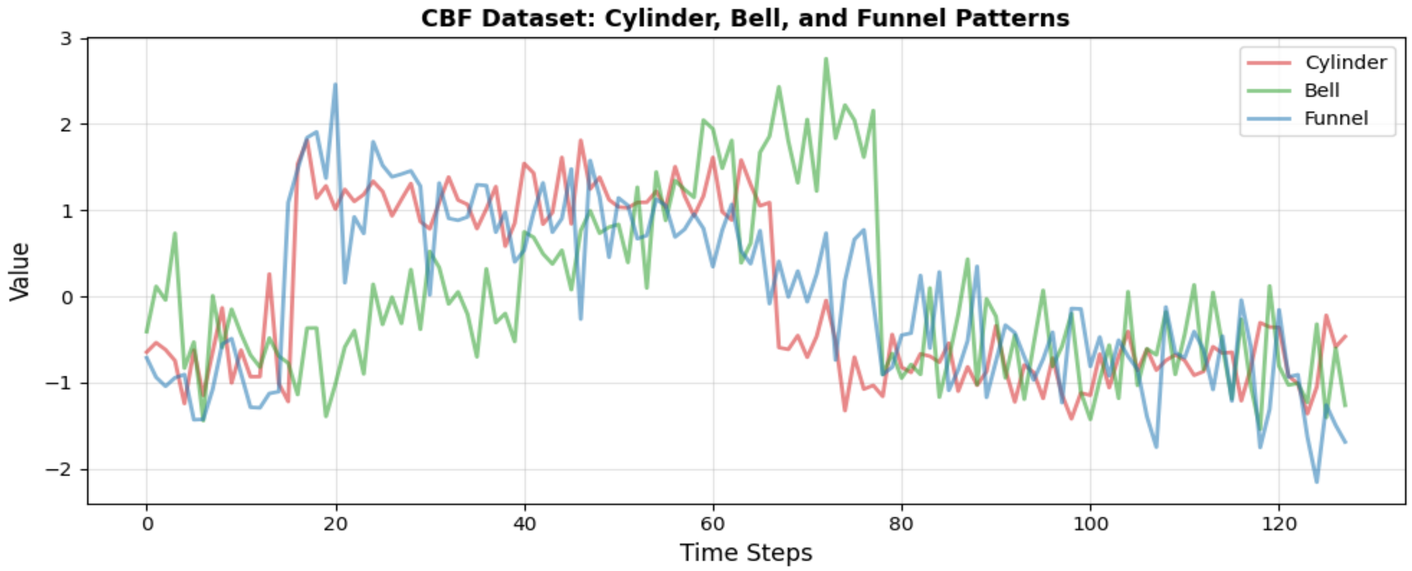


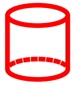


- Global faithfulness remains largely unchanged across merge percentiles.
- **p = 95** provides the most compact explanations without sacrificing faithfulness.

- Increasing merge percentile progressively merges similar event clusters.
- The number of global clusters decreases substantially.

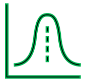
CBF Dataset – Shape-Based Time Series Classification





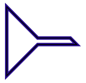
Cylinder

- Plateau bounded by two transitions.



Bell

- Gradual rise followed by a sharp decline.



Funnel

- Sharp rise followed by a gradual decline.

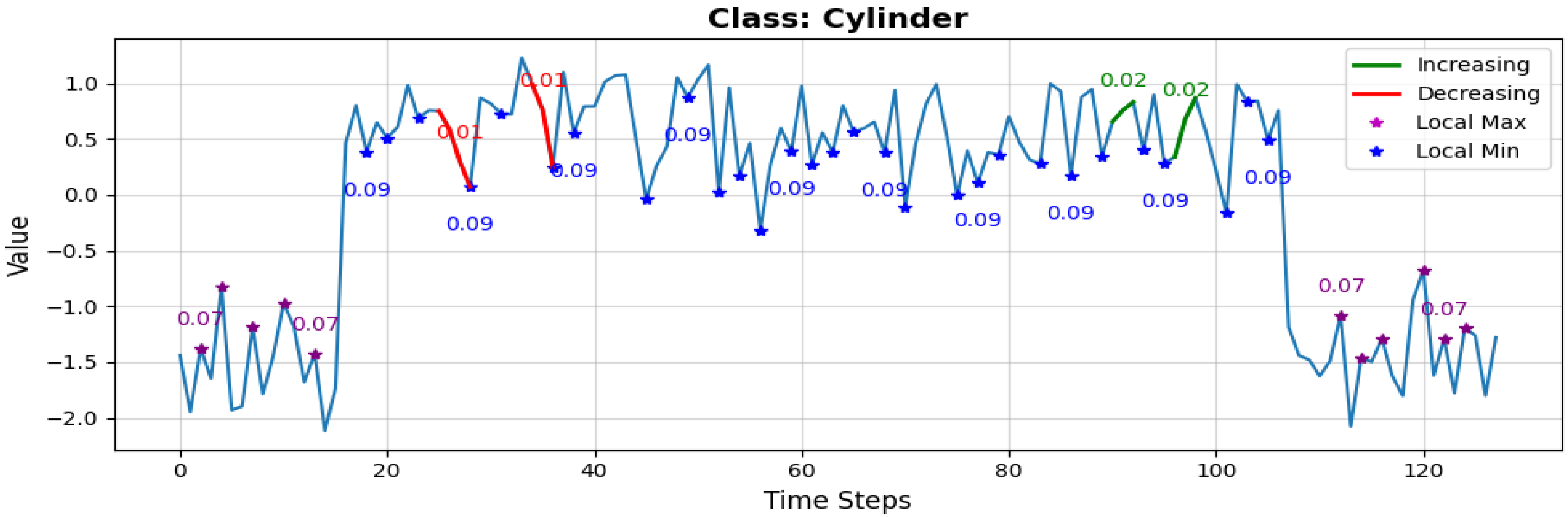
CBF

- Synthetic benchmark with known temporal patterns.
- Know class structure enables to evaluate whether explanations recover class-defining structures.

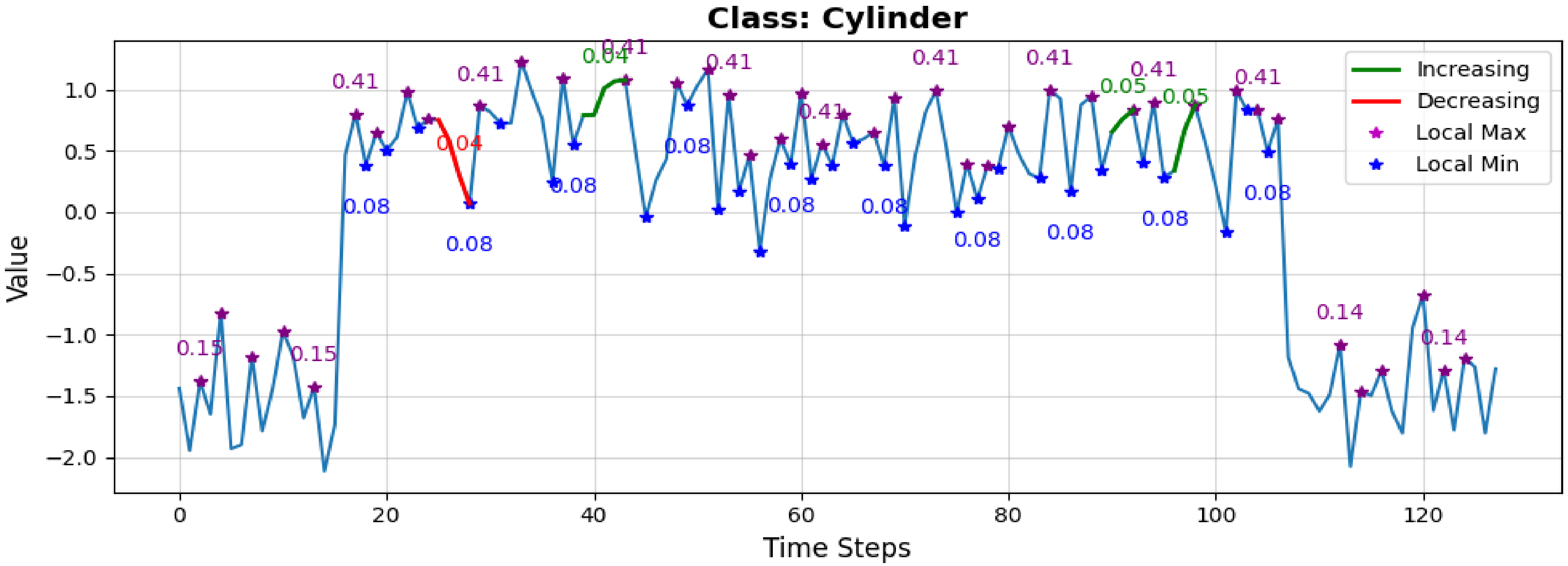
CASE STUDY

CBF: Local Explanations - Cylinder

FCN



LSTM-FCN

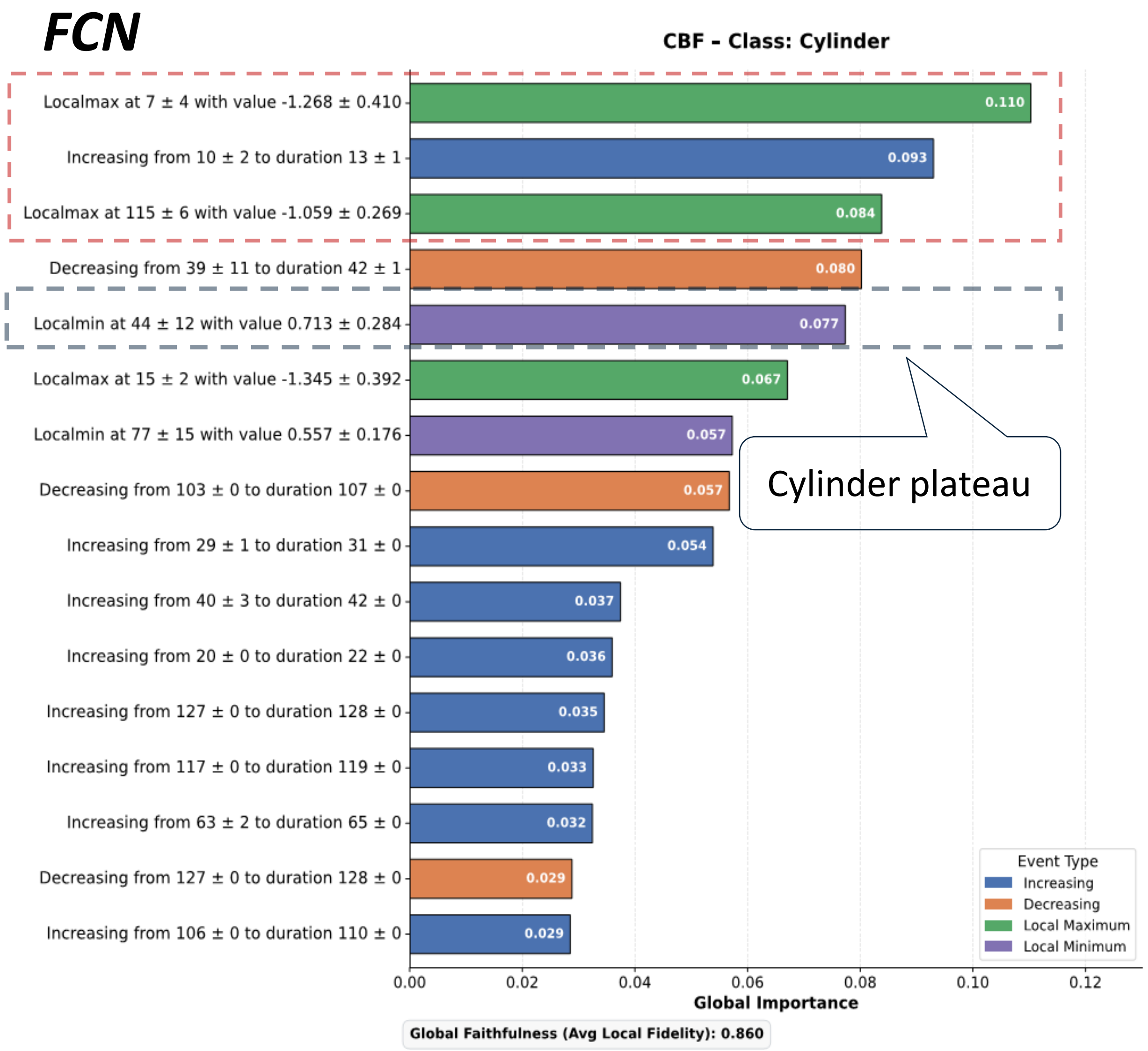


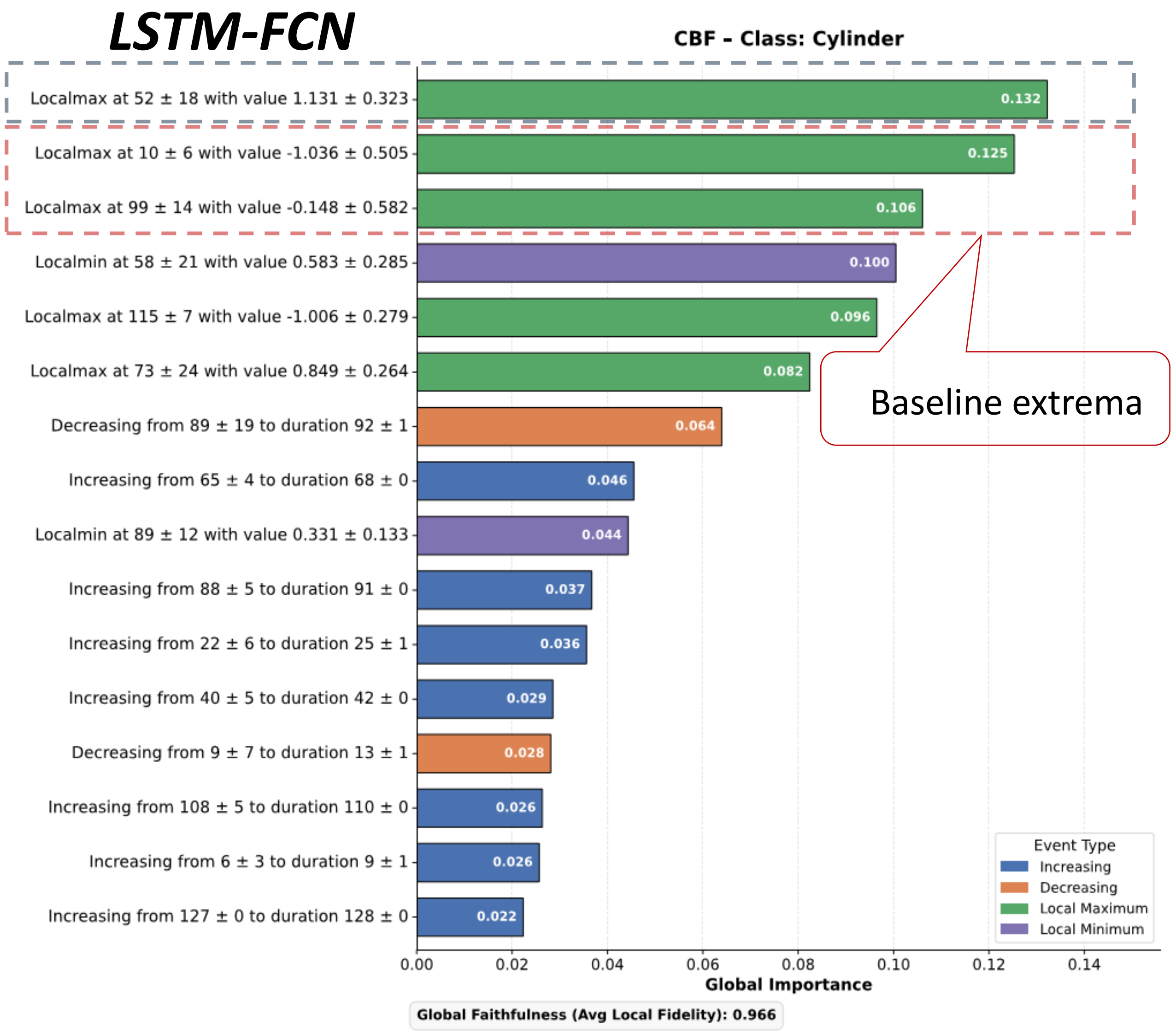
- Important events occur around the plateau region and its boundaries.

- Extrema and transitions contribute strongly to the prediction.

- The explanation is consistent with the expected Cylinder morphology.

CBF: L2GTX Global Explanations - Cylinder



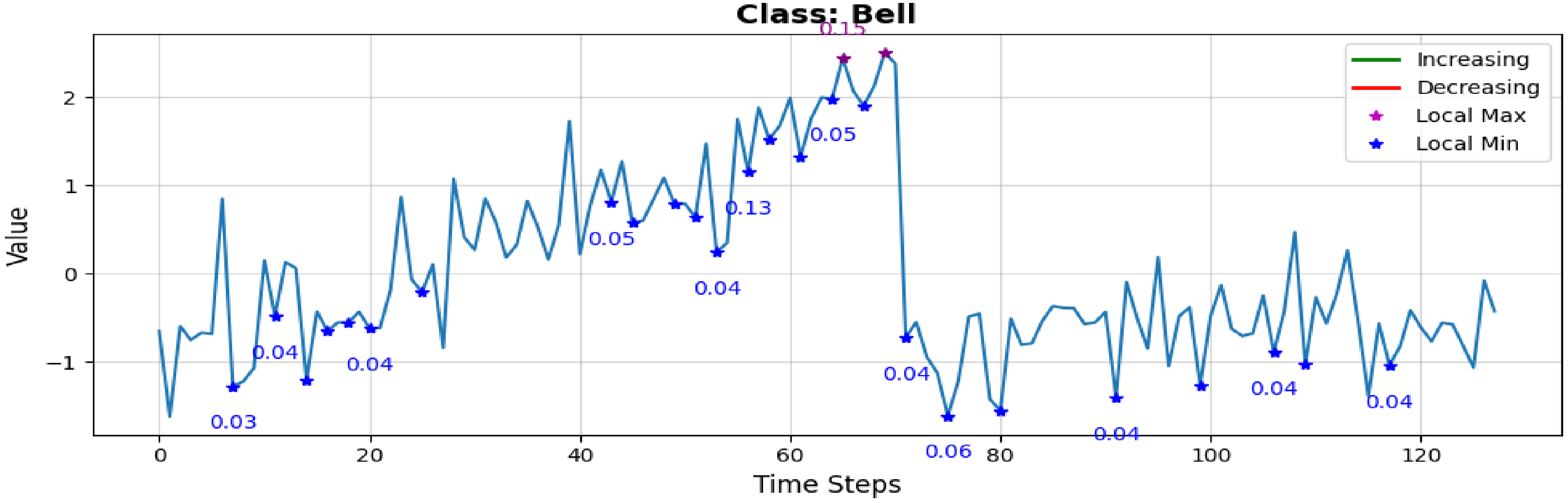


- The most important events are concentrated around the plateau and surrounding baseline regions.

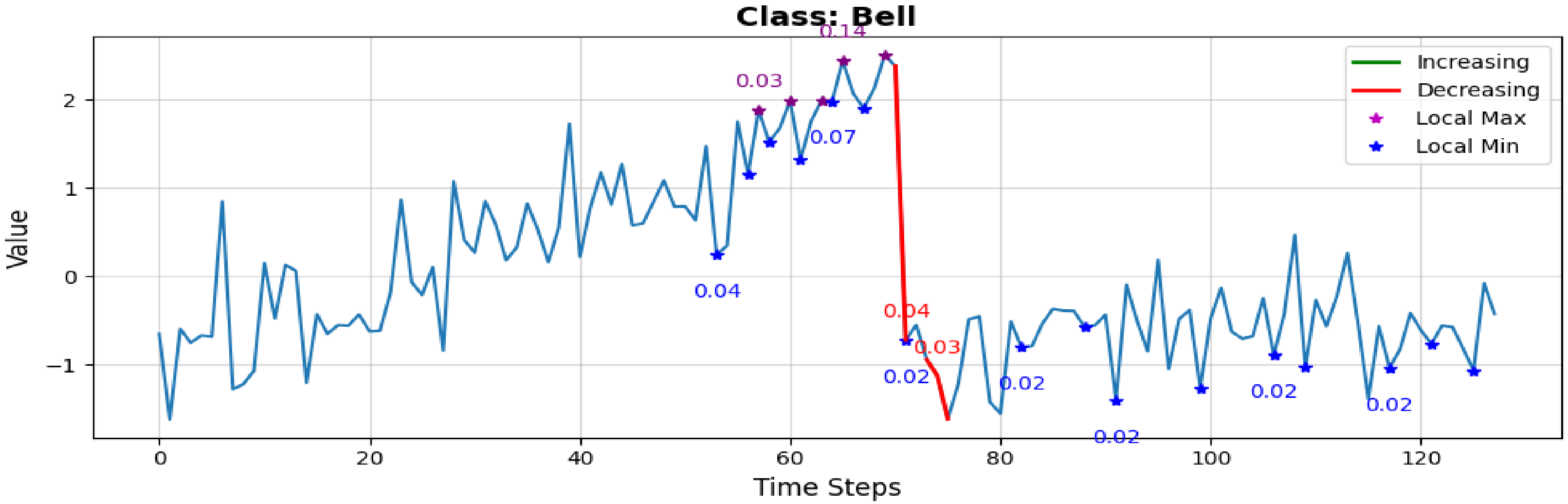
CASE STUDY

CBF: Local Explanations - Bell

FCN



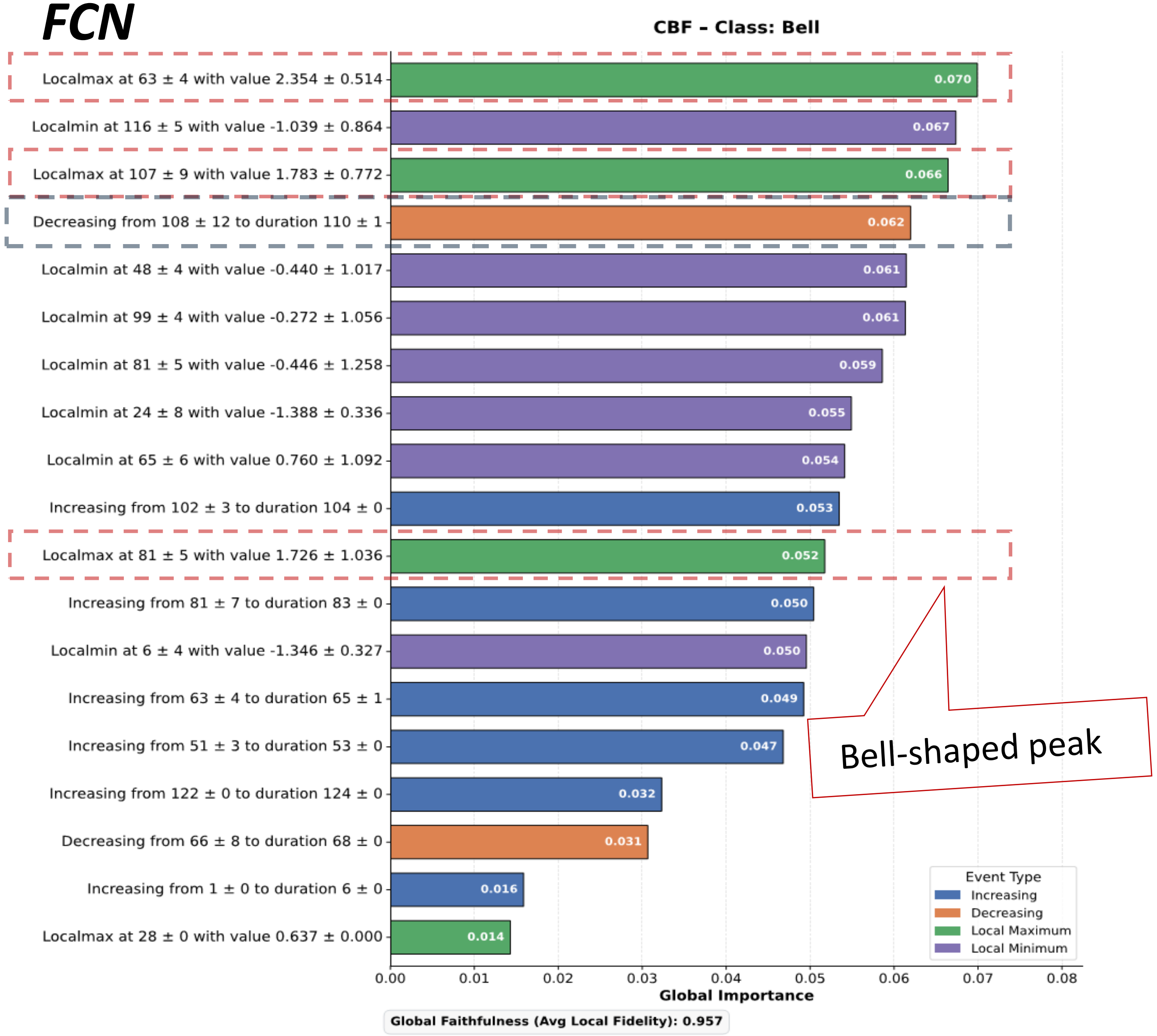
LSTM-FCN



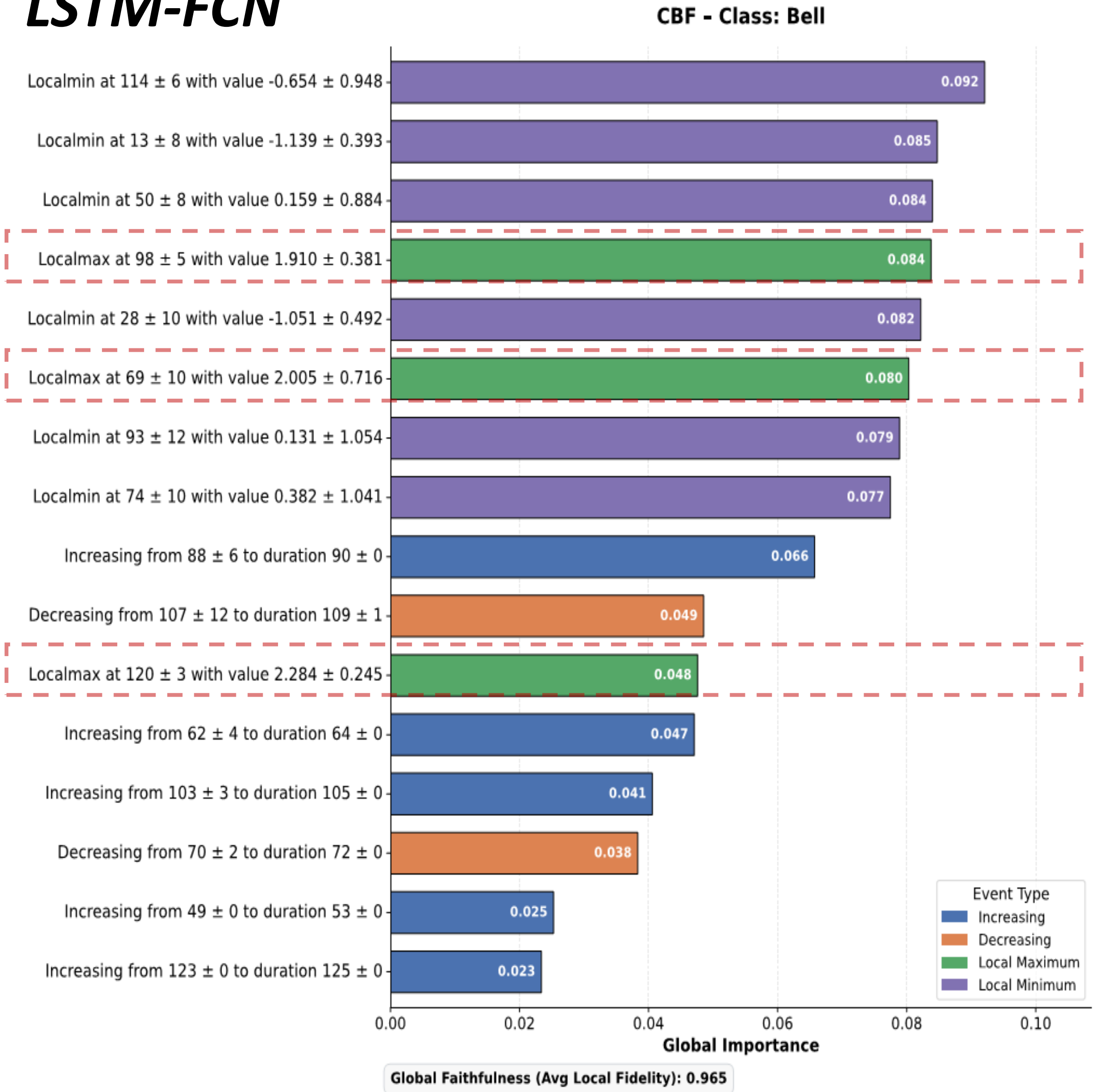
- Important events are concentrated around the peak region.
- Local extrema capture the characteristic Bell-shaped peak.
- Both models identify extrema around the peak region.

CBF: L2GTX Global Explanations - Bell

FCN



LSTM-FCN

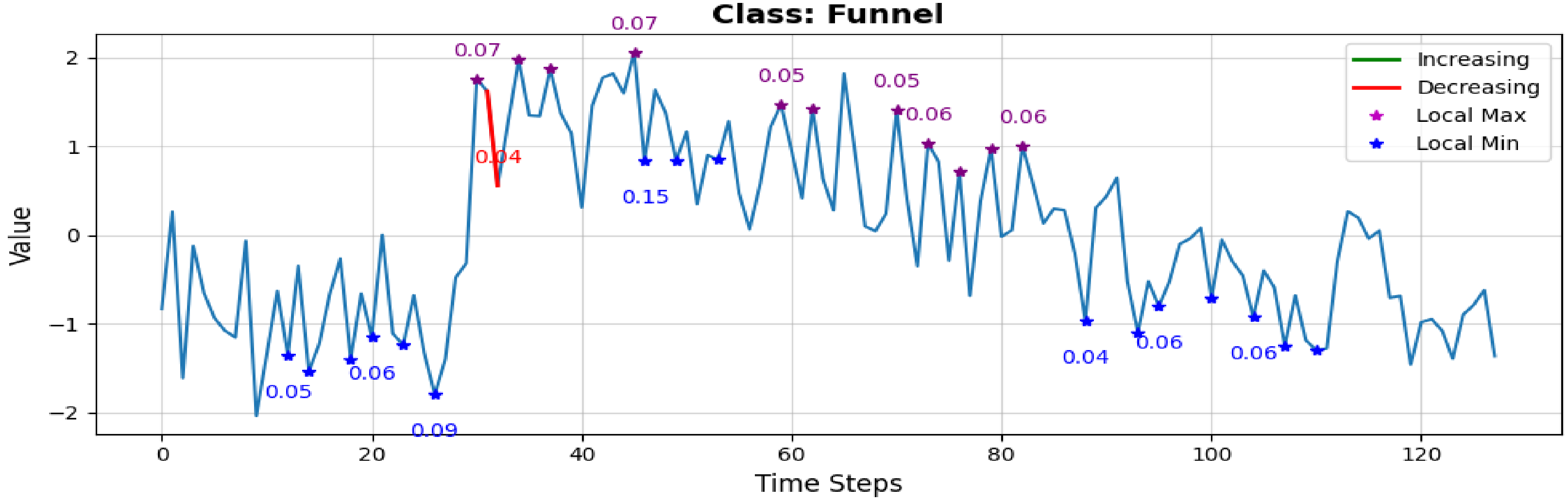


- The most important events are concentrated around the characteristic Bell-shaped peak structure.

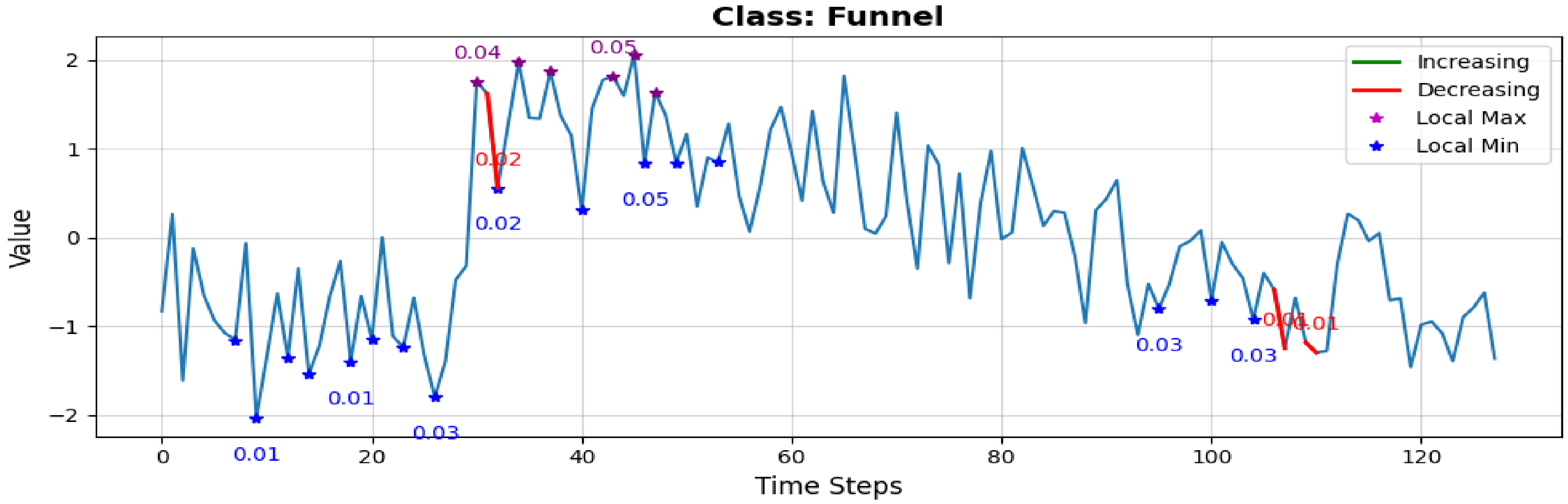
CASE STUDY

CBF: Local Explanations - Funnel

FCN



LSTM-FCN



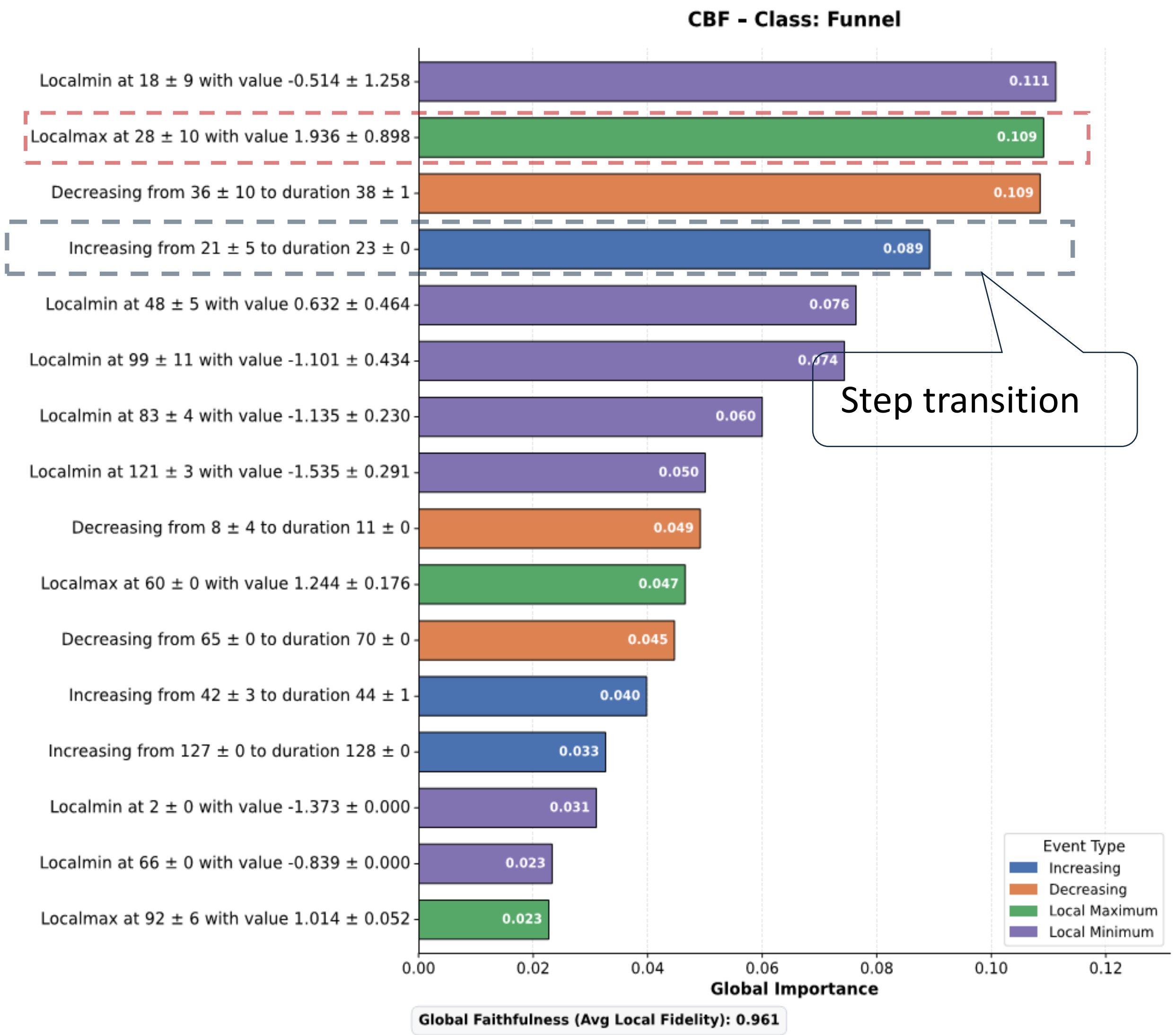
- Important events occur around the transition into the high-value region.

- Both models identify the rise transition and neighbouring peaks.

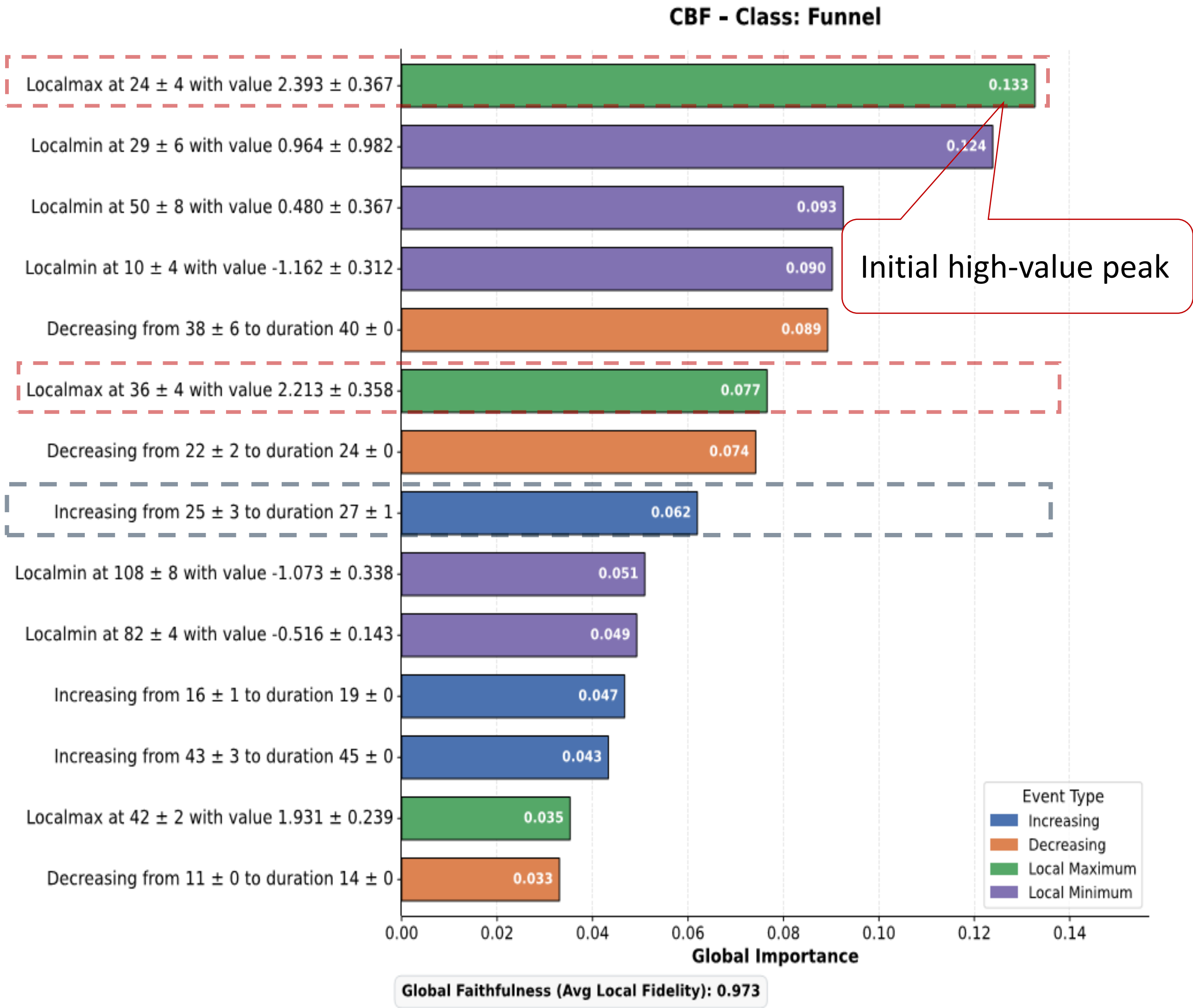
CASE STUDY

CBF: L2GTX Global Explanations - Funnel

FCN

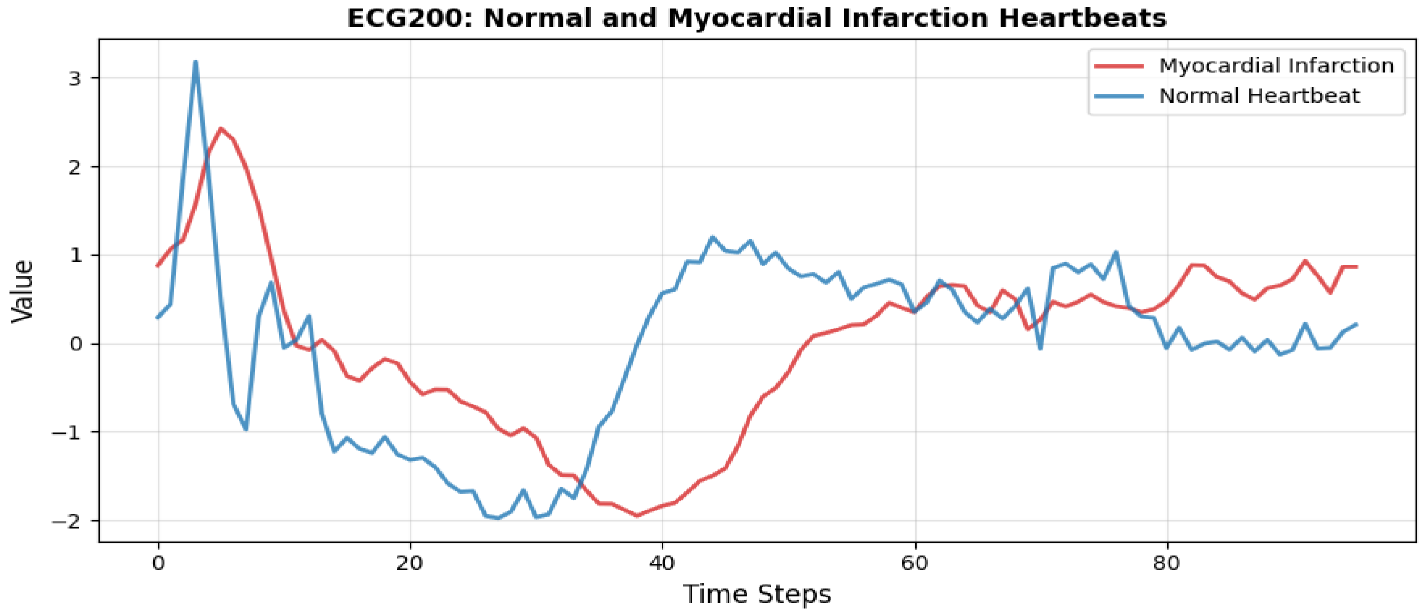


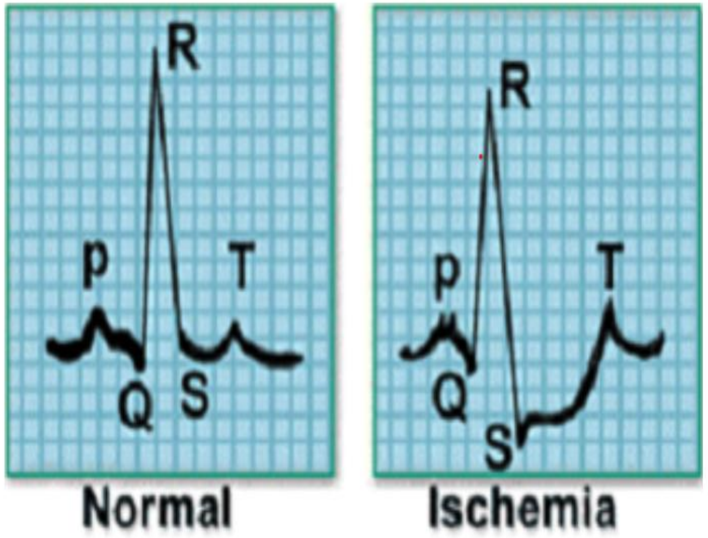
LSTM-FCN



- The most important events are concentrated around the rise transition, leading peak, and subsequent decline.

ECG200: Dataset Overview





Clinical ECG illustration comparing normal and ischemic waveforms.
Source: UCR Time Series Classification Archive.

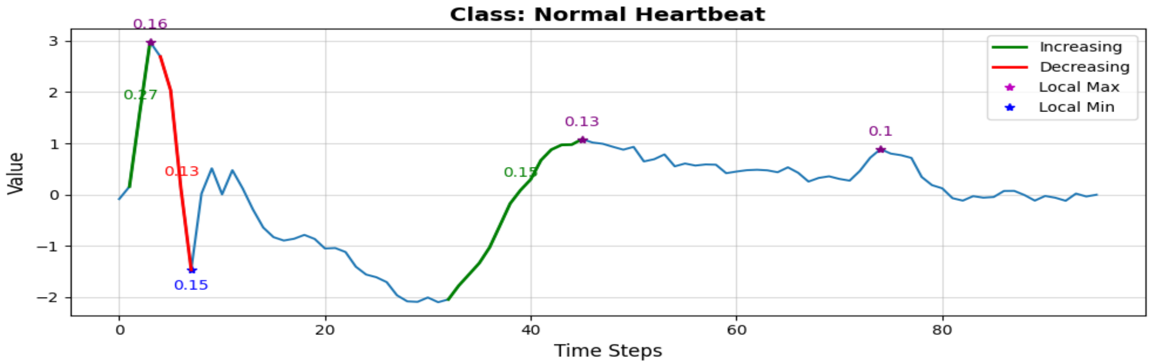
ECG200

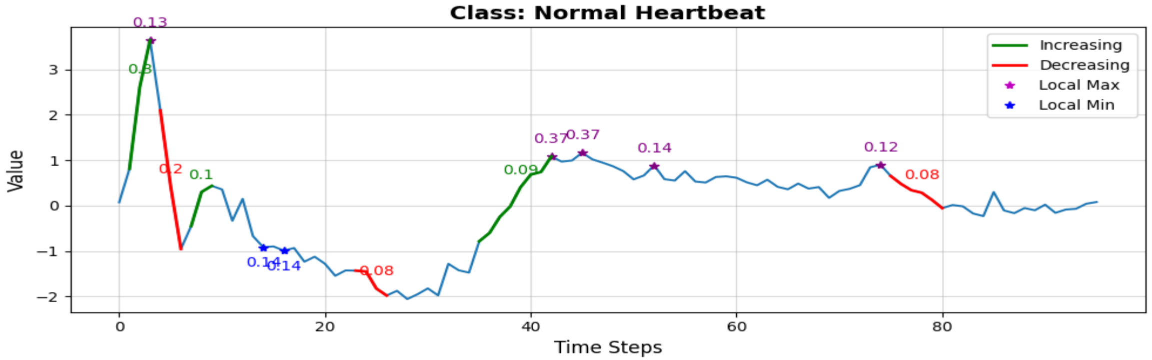
- Real-world ECG classification dataset.
- Distinguishes Normal Heartbeats and Myocardial Infarction (MI) heartbeats.
- Used to assess whether explanations capture clinically meaningful waveform differences.

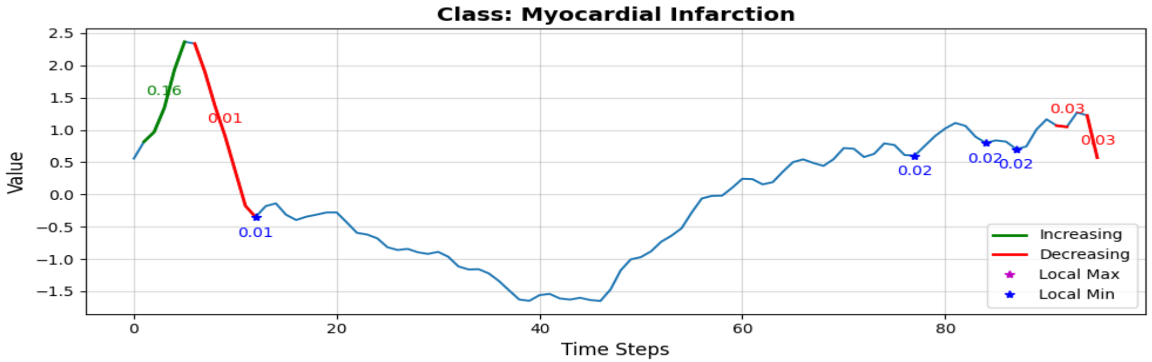
CASE STUDY

ECG200: Local Explanations

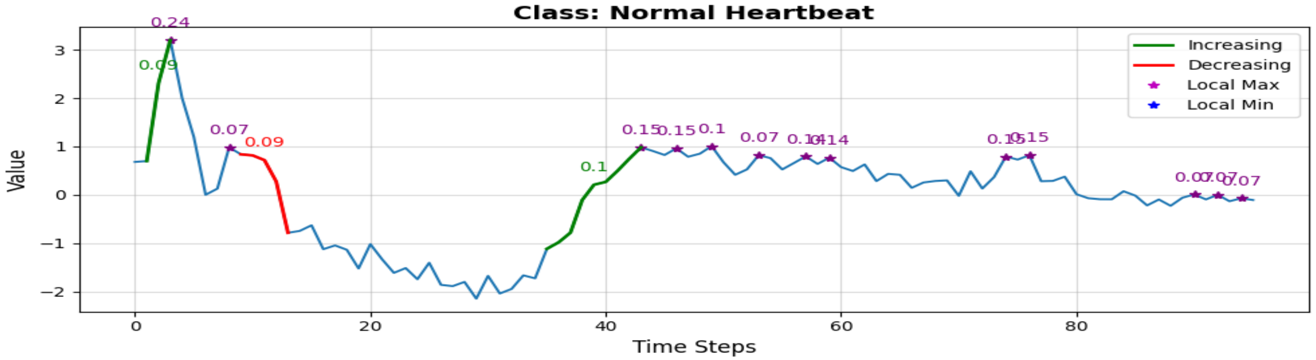
FCN

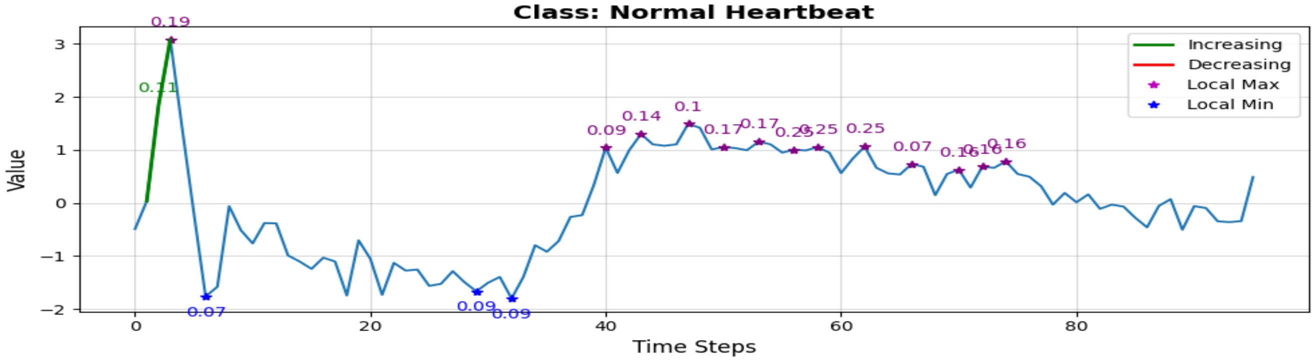


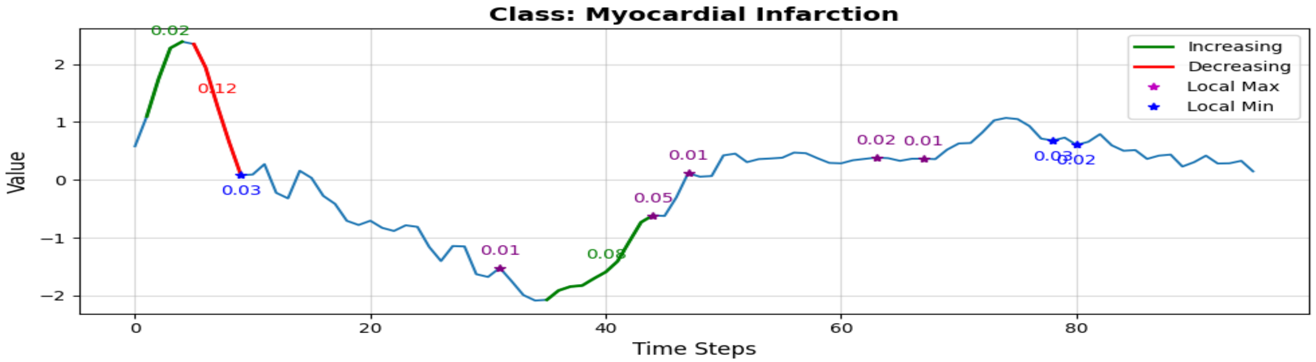




LSTM-FCN



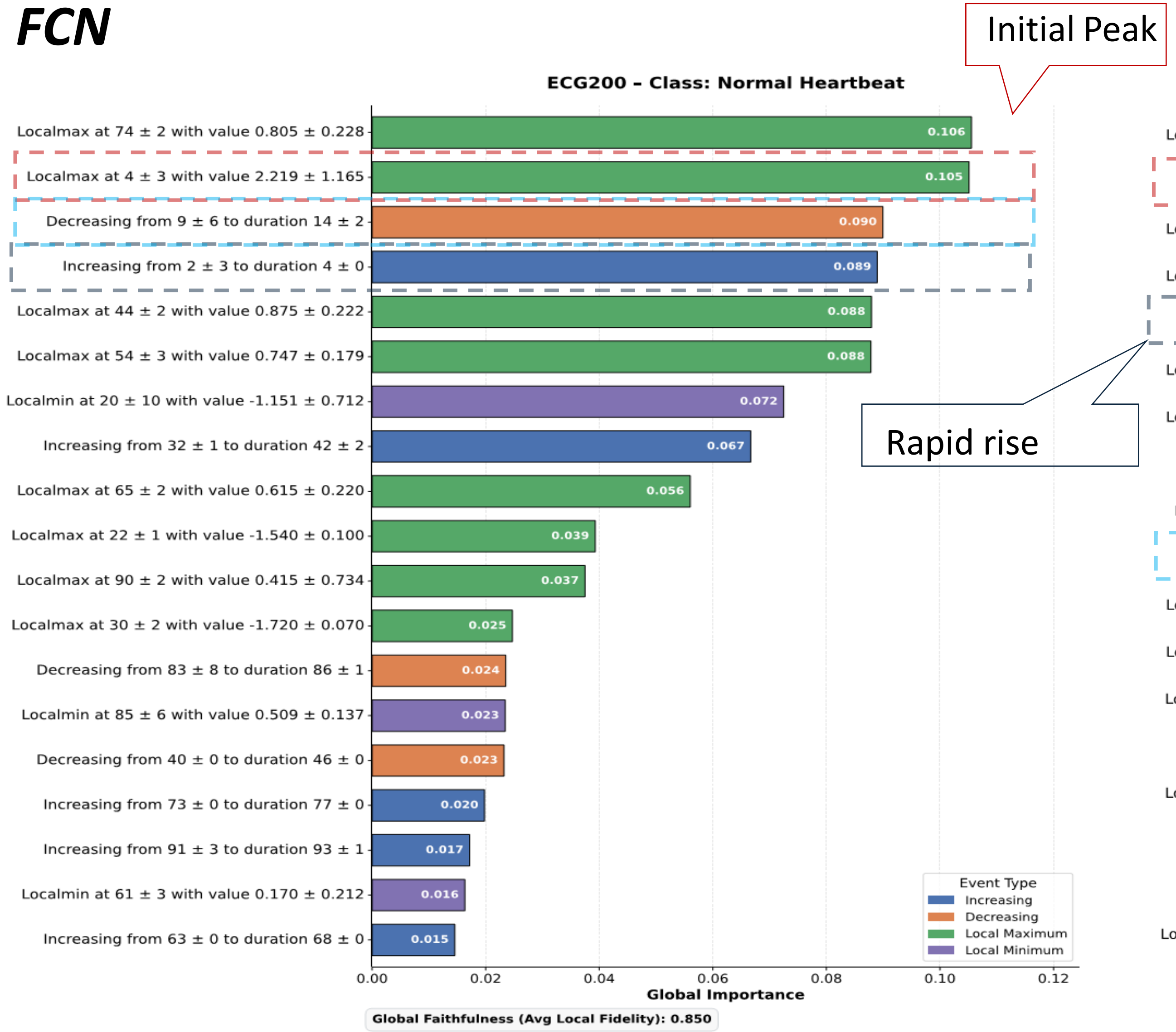




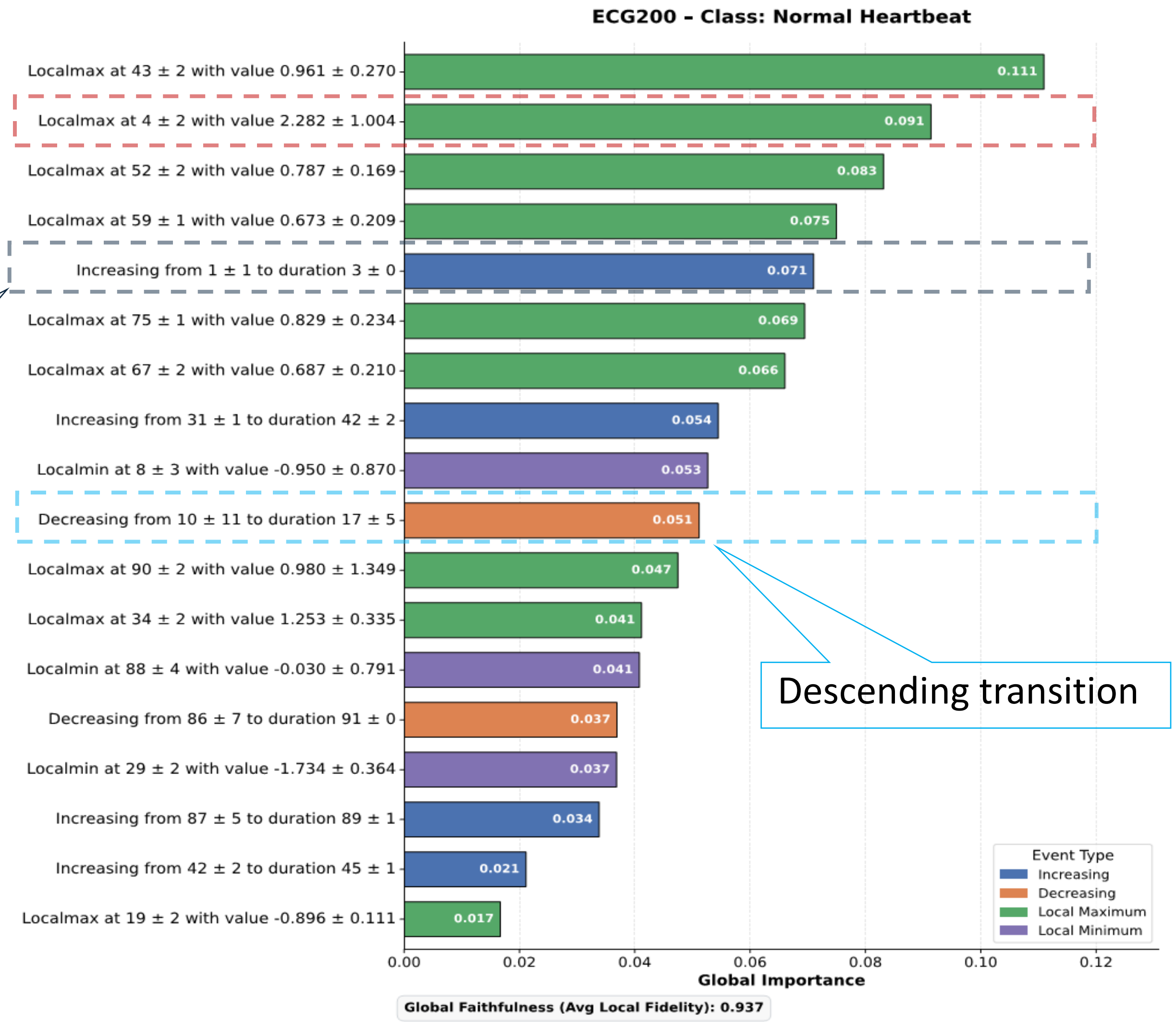
CASE STUDY

ECG200: L2GTX Global Explanations - Normal Heartbeat

FCN



LSTM-FCN

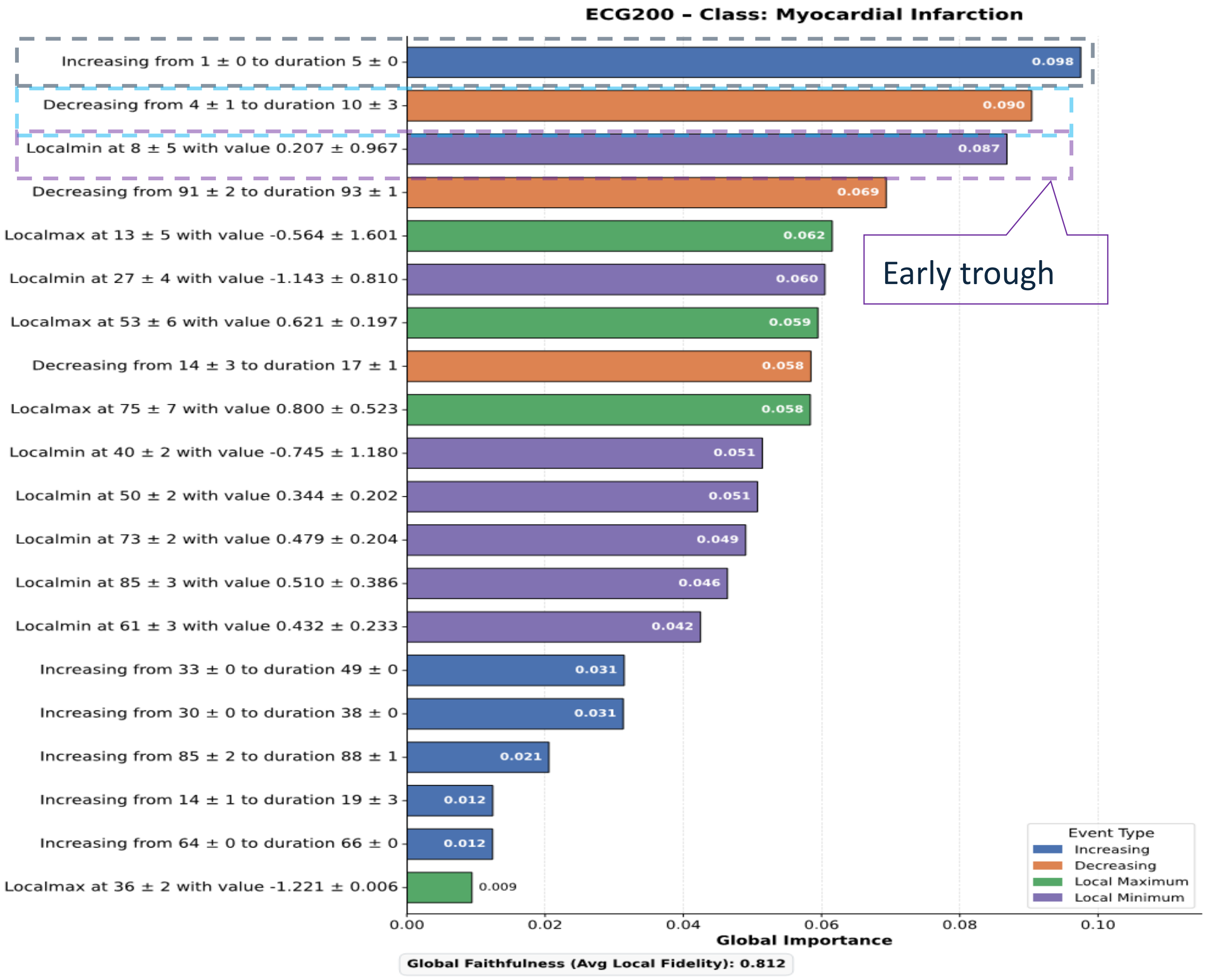


- The most important events capture the dominant initial peak and surrounding transitions.

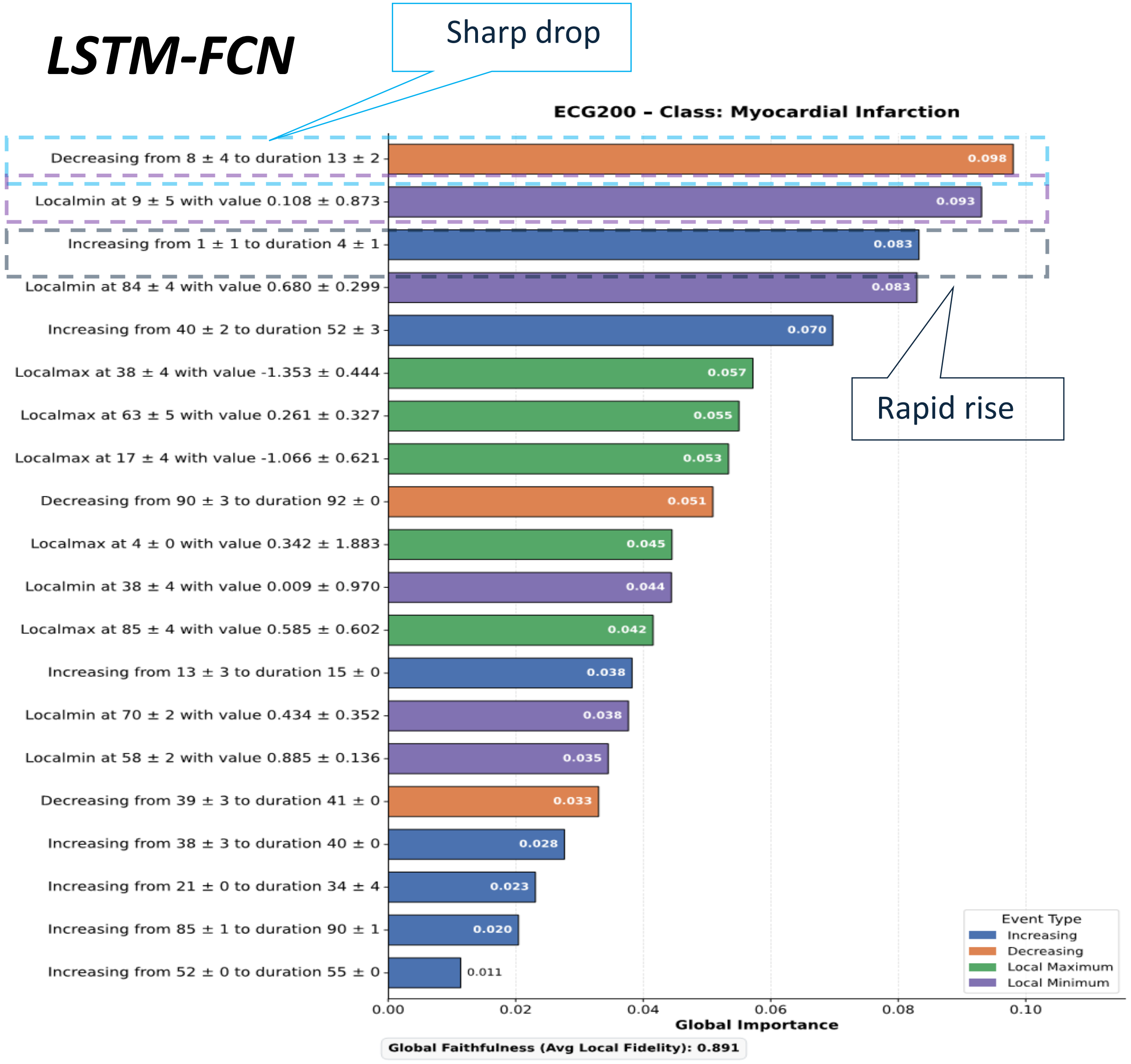
CASE STUDY

ECG200: L2GTX Global Explanations - Myocardial Infarction

FCN



LSTM-FCN



- The most important events are concentrated around the rapid rise, sharp decline, and early trough.

Contribution & Key Takeaway

Main Contribution

- L2GTX: A model-agnostic local-to-global framework for generating class-wise global explanations of time-series classifiers.

Key Characteristics

✓ Class-wise explanations ✓ Temporal-aware ✓ Human-readable ✓ Model-agnostic

Key Takeaway

- L2GTX enables model-agnostic synthesis of class-wise global explanations from local explanations while preserving temporal structure.

Future Directions

- Extend to multivariate time series data.
- Human-centred evaluation with domain experts.
- Improve scalability of event-level clustering for resource-constrained environments.

Thank You!

Questions?

Paper

Code

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